

The Scenario: The operations team believes ridership is driven by weather and by the rhythm of the week, but nobody has ever put the numbers together. Your manager wants one table: one row per day, showing how many rides happened, how long they lasted on average, and what the weather was like. “Get me that table,” she says, “and in Part 2 we’ll see if we can actually predict demand.”

```
SELECT *  
FROM `bigquery-public-data.new_york_citibike.citibike_trips`  
LIMIT 10
```

3 Tables in this TLAB: ! **Note the asterisk in gsod20*.**

Table 1 — citibike_trips - `bigquery-public-data.new\_york\_citibike.citibike\_trips`

Table 2 — noaa_gsod.gsod20* - `bigquery-public-data.noaa\_gsod.gsod20\*`

Table 3 — noaa_gsod.stations - `bigquery-public-data.noaa\_gsod.stations`

Step 1: Build Repo - https://github.com/pbcheerio/CITIBIKE_RIDERSHIP_MODEL_3

Step 2: Find Weather Station - usaf_id = 725030

```
SELECT usaf, name, country, state  
FROM `bigquery-public-data.noaa_gsod.stations`  
WHERE name LIKE '%LA GUARDIA%' AND country = 'US';
```

Query results				
Job information		Results	Visualization	JSON
		Execution details	Execution graph	
Row	usaf	name	country	state
1	725030	LA GUARDIA AIRPORT	US	NY

Step 3: Preview Ride Data

```
SELECT starttime, stoptime, tripduration  
FROM `bigquery-public-data.new_york_citibike.citibike_trips`  
LIMIT 10
```

>|

Query results

Job information		Results	Visualization	JSON	Execution details	Execution graph
Row	starttime	stoptime	tripduration			
1	2016-03-21T14:12:58	2016-03-21T14:21:18	500			
2	2013-08-22T12:49:11	2013-08-22T13:03:38	867			
3	2018-01-30T20:34:22.523000	2018-01-30T20:44:24.241000	601			
4	2017-08-28T12:26:27	2017-08-28T12:33:54	447			
5	2018-02-08T07:49:17.444000	2018-02-08T07:59:09.621000	592			
6	2013-09-20T16:39:30	2013-09-20T17:05:01	1531			
7	2018-05-10T15:45:00.867000	2018-05-10T15:50:28.711000	327			
8	2014-04-28T17:37:14	2014-04-28T17:51:07	833			
9	2016-03-24T13:27:38	2016-03-24T13:38:03	625			
10	2015-03-19T07:45:00	2015-03-19T07:49:00	220			

Step 4: Size up the Problem - total_trips = 58,937,715

```
SELECT COUNT(*) AS total_trips
FROM `bigquery-public-data.new_york_citibike.citibike_trips`;
```

>| Query results

Job information

Results

Visualization

JSON

Execution details

Execution graph

Row

total_trips

1

58937715

Step 5: Rides Per Day

```
SELECT DATE(starttime) AS ride_date, COUNT(*) AS num_rides
FROM `bigquery-public-data.new_york_citibike.citibike_trips`
WHERE starttime IS NOT NULL
GROUP BY ride_date ORDER BY ride_date;
```

Query results			
Job information		Results	Visualization
JSON		Execution details	
Execution graph			
Row	ride_date	num_rides	
1	2013-07-01	16650	
2	2013-07-02	22745	
3	2013-07-03	21864	
4	2013-07-04	22326	
5	2013-07-05	21842	
6	2013-07-06	20467	
7	2013-07-07	20477	
8	2013-07-08	21615	
9	2013-07-09	26641	
10	2013-07-10	25732	

Query results			
Job information		Results	Visualization
JSON		Execution details	
Execution graph			
Row	ride_date	num_rides	
1601	2018-05-22	48508	
1602	2018-05-23	72655	
1603	2018-05-24	74810	
1604	2018-05-25	69345	
1605	2018-05-26	51597	
1606	2018-05-27	25814	
1607	2018-05-28	49043	
1608	2018-05-29	69090	
1609	2018-05-30	72321	
1610	2018-05-31	55794	

Step 6: Add Avg Trip Duration

```
SELECT DATE(starttime) AS ride_date, COUNT(*) AS num_rides,
ROUND(AVG(tripduration) / 60, 2) AS avg_duration_min
FROM `bigquery-public-data.new_york_citibike.citibike_trips`
WHERE starttime IS NOT NULL
GROUP BY ride_date ORDER BY ride_date;
```

>|

Query results

Job information		Results	Visualization	JSON	Execution details	Execution graph
Row	ride_date	num_rides	avg_duration_min			
1	2013-07-01	16650	16.31			
2	2013-07-02	22745	15.97			
3	2013-07-03	21864	16.24			
4	2013-07-04	22326	21.22			
5	2013-07-05	21842	18.04			
6	2013-07-06	20467	19.33			
7	2013-07-07	20477	18.64			
8	2013-07-08	21615	21.49			
9	2013-07-09	26641	16.1			
10	2013-07-10	25732	15.53			

Step 7: One Station, 6 Years of Weather

```
SELECT
  PARSE_DATE('%Y-%m-%d', CONCAT(year, '-', mo, '-', da)) AS obs_date,
  temp AS temp_f, `max` AS max_temp_f, `min` AS min_temp_f,
  wdsp AS wind_speed_knots, prcp AS precip_in
FROM `bigquery-public-data.noaa_gsod.gsod20*`
WHERE _TABLE_SUFFIX BETWEEN '13' AND '18' AND stn = '725030';
```

Query results							
Job information		Results	Visualization	JSON	Execution details	Execution graph	
Row	obs_date	temp_f	max_temp_f	min_temp_f	wind_speed_knots	precip_in	
1	2018-01-12	47.9	60.1	35.1	9.4	0.01	
2	2018-09-12	73.8	78.1	64.0	4.9	0.05	
3	2018-12-15	47.9	52.0	39.9	5.4	0.23	
4	2018-06-04	59.0	70.0	51.8	10.2	0.64	
5	2018-01-30	34.2	44.1	30.0	11.2	0.06	
6	2018-04-04	45.9	57.0	39.0	9.3	0.28	
7	2018-03-30	50.1	60.1	43.0	6.3	0.02	
8	2018-03-07	36.6	41.0	33.8	14.7	0.1	
9	2018-07-23	79.5	86.0	72.0	13.4	0.4	
10	2018-11-10	47.8	54.0	39.9	14.7	0.74	

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Step 8: Join - Final Query

```
WITH daily_rides AS (
  SELECT
    DATE(starttime) AS ride_date, COUNT(*) AS num_rides,
    ROUND(AVG(tripduration) / 60, 2) AS avg_duration_min
  FROM `bigquery-public-data.new_york_citibike.citibike_trips`
  WHERE starttime IS NOT NULL
  GROUP BY ride_date
```

```

),
daily_weather AS (
  SELECT
    PARSE_DATE('%Y-%m-%d', CONCAT(year, '-', mo, '-', da)) AS obs_date,
    temp AS temp_f, `max` AS max_temp_f, `min` AS min_temp_f,
    wdsp AS wind_speed_knots, prcp AS precip_in
  FROM `bigquery-public-data.noaa_gsod.gsod20*`
  WHERE _TABLE_SUFFIX BETWEEN '13' AND '18' AND stn = '725030'
)
SELECT
  r.ride_date, r.num_rides, r.avg_duration_min,
  w.temp_f, w.max_temp_f, w.min_temp_f,
  w.wind_speed_knots, w.precip_in,
  FORMAT_DATE('%A', r.ride_date) AS day_of_week,
  EXTRACT(MONTH FROM r.ride_date) AS month
FROM daily_rides AS r
INNER JOIN daily_weather AS w
  ON r.ride_date = w.obs_date
ORDER BY r.ride_date;

```

Query results										Chat	View all jobs	Save results
Job information		Results	Visualization	JSON	Execution details		Execution graph					
Row	ride_date	num_rides	avg_duration_min	temp_f	max_temp_f	min_temp_f	wind_speed_knots	precip_in	day_of_week			
1	2013-07-01	16650	16.31	74.8	78.1	73.4	7.8	0.0	Monday			
2	2013-07-02	22745	15.97	76.1	82.9	73.0	8.0	0.73	Tuesday			
3	2013-07-03	21864	16.24	78.5	84.9	73.9	8.8	0.06	Wednesday			
4	2013-07-04	22326	21.22	82.0	91.0	73.9	8.6	0.96	Thursday			
5	2013-07-05	21842	18.04	84.4	93.0	75.9	9.0	0.0	Friday			
6	2013-07-06	20467	19.33	87.0	95.0	77.0	9.3	0.0	Saturday			
7	2013-07-07	20477	18.64	88.8	98.1	79.0	8.8	0.0	Sunday			
8	2013-07-08	21615	21.49	83.7	98.1	75.0	8.9	0.0	Monday			
9	2013-07-09	26641	16.1	82.6	91.0	75.0	6.0	0.0	Tuesday			
10	2013-07-10	25732	15.53	82.0	91.0	75.0	8.7	0.08	Wednesday			

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Step 9: Export and Commit - Done

https://github.com/pbcheerio/CITIBIKE_RIDERSHIP_MODEL_3